Solar System Models · Teacher and Cover Plan

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Scheme Outcome

TRACK Science scheme of work: Earth and Space

Use solar system data and a scaled model to investigate the relationship between a planet's distance from the Sun and its orbit time, and evaluate model limitations.

Success Criteria

I can read planet distance and orbit time from a solar system data table.

I can describe the pattern linking distance from the Sun to the time for one orbit.

I can state one strength and one limitation of a classroom solar system model.

Model Sequence (I do)

First, look at a 1-metre paper strip model: the four rocky planets cluster in the first 5 cm near the Sun!

Second, the four giant planets spread out across the rest of the strip all the way to 100 cm (Neptune).

Third, check the data table: every step further from the Sun gives a longer orbit time.

Our 1-metre strip model is great for showing how bunched up the inner planets are compared with the outer giants.

Made by Matt

However, if we shrink the solar system to 1 metre, Earth itself would be smaller than a speck of dust!

Also, in real space the planets do not sit in a frozen straight line·they are constantly moving at different speeds.

We Do & Hinge Check

Saturn is further from the Sun than Jupiter (12 years) and closer than Uranus (84 years). Which value in the table is Saturn's orbit time?

Answer: A. Following the pattern, Saturn's orbit time (29 Earth years) sits between Jupiter (12 years) and Uranus (84 years).

Safe Practical & Card Setup

Mark the scaled planet positions on a 100 cm paper strip and compare the orbit data table.

Kit: 100 cm paper strip, metre rule, pencil, and printed Solar System Data Table.

Keep metre rules flat on the desk when marking the paper strip.

You Do Routes (Supported, Independent, Aspire)

Supported: Put Mercury, Earth, Jupiter, and Neptune in order from shortest orbit to longest orbit and finish the pattern sentence. (Help: Closest = Mercury (88 days). Furthest = Neptune (165 years). Further from the Sun = __ orbit time.)

Independent: Use the data table to state the pattern linking distance from the Sun to orbit time (quoting

two planets' numbers), and explain one strength and one limitation of the strip model. (Help: As distance from the Sun increases from __ (Mercury) to __ (Neptune), orbit time increases from __ to __. One limitation of the strip model is __.)

Aspire: Explain why distance and planet size cannot share the same scale on a 1 m strip, and note how a planet's atmosphere (like Venus's thick carbon dioxide clouds) can affect surface temperature. (Help: On a 1 m scale where 4,500 million km = 100 cm, Earth's 12,742 km diameter would be less than __ mm. Also, while orbit time follows distance strictly, temperature can be affected by __.)